

Spectral discrimination of an invasive species (*Spartina alterniflora*) in Min river wetland using field spectrometry

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ABSTRACT

Spartina alterniflora, an invasive plant, has been a threat to the local ecological security since it was introduced to Fujian coastal beach over 30 years ago. How to monitor its dynamic changes effectively is of great significance. Currently, hyperspectral remote sensing technology has become an important way to monitor invasive species dynamic changes. This paper investigates whether *S. alterniflora* could be discriminated from the other three native species using field spectrometer ranging from 350 nm to 2500 nm. In order to reduce and select the optimal bands for the potential discrimination of *S. alterniflora*, a hierarchical method is implemented to spectrally discriminate *S. alterniflora* from the other three native species. In the first level of the analysis using ANOVA, we found that there were statistically significance differences in spectral reflectance between *S. alterniflora* and the other three native species at different bands. The algorithm of classification and regression trees (CART) were used to further investigate in the second level of analysis to identify the most sensitive bands for spectral discrimination. We found that the greatest discrimination power for *S. alterniflora* is located in the red-edge, especially in the near infrared, and mid infrared regions. Subsequently, we used Jeffries-Matusita (JM) distance to assess spectral separability of bands selected by CART. Overall, results of this study offer the possibility of extending field measurements at canopy level to airborne and hyperspectral data for discriminating *S. alterniflora* in Min river wetland.

Keywords: Spectral reflectance, CART, Jeffries-Matusita distance, ANOVA, invasive species, native species

1. Introduction

Spartina alterniflora is an invasive species of special concern in the coastal wetlands of China, which was recognized as one of 16 most harmful invasive exotic species by the State Environmental Protection Administration (SEPA) of China in 2003^[1]. The primary concern relative to *S. alterniflora* invasion is the loss of species richness and consequent potential for extinction and loss of biodiversity^[2, 3].

To understand the magnitude and distribution of invasive species, systematic, comprehensive monitoring programs are needed. Traditional methods of monitoring invasive species infestations are costly, time consuming, and often require intensive field work^[4]. However, remote sensing offers a practical and economical means to monitor and discriminate it from other wetland species. Many researchers have used both multi-spectral data such as QuickBird and SPOT imagery

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to monitoring the dynamic of invasion species^[5, 6], and hyperspectral data to discriminate and map invasive wetland vegetation^[7], and promising results have been obtained. But there are lots of challenges, which lead to difficult to detectable as terrestrial plants^[4].

Hyperspectral remote sensing has hundreds of narrow continuous spectral bands between 350 and 2500 nm, which can improve species discrimination by capitalizing on the biochemical and structural vegetation properties^[8, 9]. However, the greater spectral dimensionality leads to data redundancy, when using, need a lot of denoising and dimensionality. The key problem is how to effective dimensionality reduction without losing any important information. Several methods can be used to reduce dimensionality, Such as principal component analysis (PCA), artificial neural network (ANN) and support vector classification (SVC)^[10, 11]. However, inconsistent results have been obtained for different species and environments^[12].

In this study, the main goal is to investigate whether field spectrometry data could be used effectively to discriminate *S. alterniflora* from native species occurring in the Min River estuary of southeast China. More specifically, the objectives of this study were: (1) to test whether there is a significant difference between the mean reflectance at each measured wavelength (from 350 to 2500 nm) for *S. alterniflora* and the other three native species. (2) To identify wavelengths that is most sensitive in discriminating *S. alterniflora* from the other three native species.

2. Methods

2.1 Study area

The study was conducted in the Shanyutan wetland (Longitude, 119°34'12"-119°40'40"; Latitude, 26°00'36"-26°03'42"), which is the largest tidal wetland in the Min River estuary, southeast China. The climate of the study area is warm and wet, with a mean annual temperature of 19.6°C and a mean annual precipitation of 1350 mm^[13]. The native dominant plant species include *Phragmites australis*, *Cyperus malaccensis* and *Kandelia candel*. They serve as an important stopover site for migratory birds on the East Asian flyway. The wetland also functions as an important sanctuary for a number of endangered species or as a spawning site for economically important aquatic species. For these reasons, a provincial natural reserve was established in 2003 in the estuary. However, local historical records showed that *S. alterniflora* began to invade this marsh in 2002 and since then this species has rapidly expanded its cover from the native species, which may threaten the estuarine ecosystems and their biodiversity.

2.2 Spectral sampling

The sampling locations were selected using field observation techniques performed in October, 2012. We choose for representative invaded mudflats and salt marshes areas. Dense canopies and stands were considered requirements for minimizing effects of soil and water background on the in situ spectral sampling.

The Analytical Spectral Devices (ASD) FieldSpec®3 spectrometer was used to measure the spectral reflectance from *S. alterniflora* and the native species. The spectrometer has a wavelength ranging from 350 to 2500 nm. The instrument has a spectral sampling resolution of 1.4 nm and a spectral interval of 3 nm between 350 to 1000 nm, and a spectral sampling resolution of 2 nm and a spectral interval of 10 nm between 1000 to 2500 nm, respectively. (FieldSpec Pro FR, Analytical Spectral Devices, Inc.)

Spectral sampling was conducted in October 2012, which was the flowering period of *P. australis* in the Min River estuary. A representative area was selected for reflectance measurements for each plant species, respectively. Spectral samples were taken every 10 m, in each sampling location, to avoid autocorrelation, at about 1.2 m above vegetation canopy, and with a 25° field of view. All spectral samples were collected under sunny, cloudless and windless conditions and between 10:00 am-02:00 pm. A white reference calibration – Spectralon panel – was used to normalize for variations in atmospheric conditions and to convert the measurements into relative reflectance ^[14]. Table 1 shows the number of spectral samples collected for *S. alterniflora* and other species.

Table 1

Number of spectral samples collected for *S. alterniflora* and the native species

Species name	Type code	Spectral samples
<i>S. alterniflora</i>	SA	51
<i>P. australis</i>	PA	44
<i>C. malaccensis</i>	CM	25
<i>K. candel</i>	KC	18

2.3 Data processing

2.3.1 Data dimensionality reduction

A hierarchical analysis was performed to reduce data dimensionality to the most sensitive wavelengths able to discriminate the *S. alterniflora* from the native species, using the approach of Adam and Mutanga ^[12]. In the first level, the differences between the species were tested for significance using the one way analysis of variance (One-way ANOVA) with the post-hoc Scheffe test at each measured wavelength for the individual class pair (SA vs. PA, SA vs. CM, SA vs. KC). We tested ANOVA with a 95% confidence level ($P < 0.05$) to ensure that no crucial wavelengths for the discrimination of vegetation types were left out.

In the second level, we used Classification and Regression Trees (CART) to further reduce the number of wavelengths previously using the significant differences bands selected by ANOVA. CART was developed by Breiman et al. ^[15], is a nonparametric statistical model that classifies the data using a top-down sequential binary splitting mechanism based on the splitting rule. CART models recursively partition the data to find increasingly homogeneous subsets based on independent variable splitting criteria using variance minimizing algorithms. The dependent data are partitioned into a series of descending left and right child nodes derived from parent nodes ^[15, 16]. Once the partitioning has ceased, the child nodes are designated as terminal nodes. Further more information about CART can be found at the Salford Systems Web site (<http://www.salford-systems.com>.) and Breiman et al ^[15].

In this study, the spectral data were divided into five spectral regions: blue (350-449 nm), green (450-549nm), red (550-649), red-edge (650-749), near infrared (750-1299 nm) and mid infrared (1300-2500 nm). We used CART to select the most sensitive wavelengths from the number of significant wavelengths selected in the first level (ANOVA) in each spectral region, since the independent variable (spectral reflectance) is continuous predictors and the dependent variable is categorical (species type). We chose the Gini index for classification trees as the loss function with a minimum proportional reduction of error at any split of 0.0001. Meanwhile, a cost -complexity pruning method was used to obtain

the optimal tree. In the end of the algorithm only the spectral bands that produce small misclassification rates, and simultaneously avoid the “overfitting” of the model, were selected.

Due to severe noises in the water absorption spectrum from 1355 to 1434 nm, 1801 to 1969 nm and 2350 to 2500 nm, only the data collected in 350 to 1354 nm, 1435 to 1800 nm, and 1970 to 2349 nm were used.

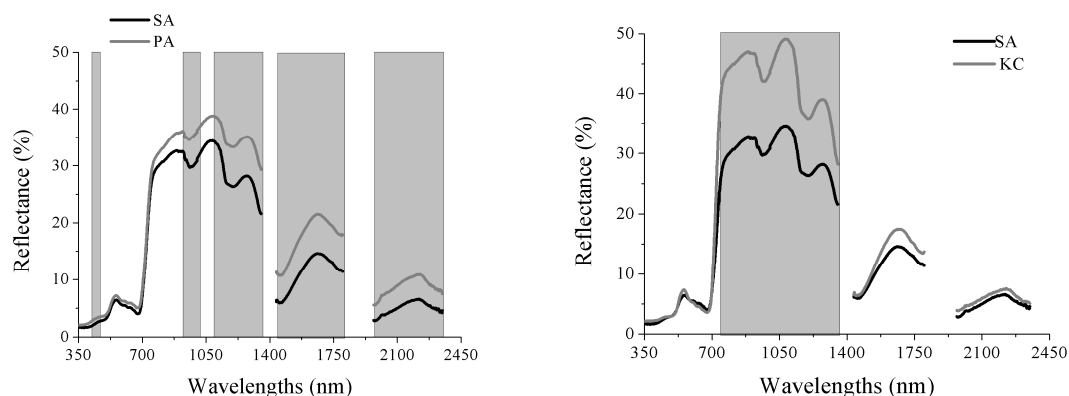
2.3.2 Spectral separability

We used Jeffries-Matusita (JM) distance to assess the spectral separability between *S. alterniflora* and other native species using the optimal primary splitter bands selected by the CART analysis. The JM distance between a pair of probability functions is seen as a quantification of the mean distance between the two class density functions^[17]. The JM distance has upper and lower bounds that vary 0 and 1.414. If the calculated JM distance is equal to the appropriate upper bound, then the signatures can be said to be totally separable of the class pairs in the band used. A calculated J-M distance of zero means that the signatures are inseparable^[18]. In this study, we used a JM distance of ≥ 1.372 ($\geq 97\%$ of 1.414) as a threshold of spectral separability between group pairs, which is even stricter than the value of 1.343 ($\geq 95\%$ of 1.414) that is commonly used in remote sensing practice^[19,20].

3. Results

3.1 ANOVA test results

The results of the ANOVA test for each class pair (SA vs. PA, SA vs. KC, and SA vs. CM) were shown in Fig. 1. The shaded areas (Fig. 1) showed the wavelengths where the spectral reflectance from the *S. alterniflora* was statistically different from the other three species, with a 95% confidence level ($P < 0.05$). Significant differences were observed in different wavelengths between *S. alterniflora* and the three native species, throughout the spectral range under analysis. These significant wavelengths are mainly located in the red-edge, near infrared and mid infrared.



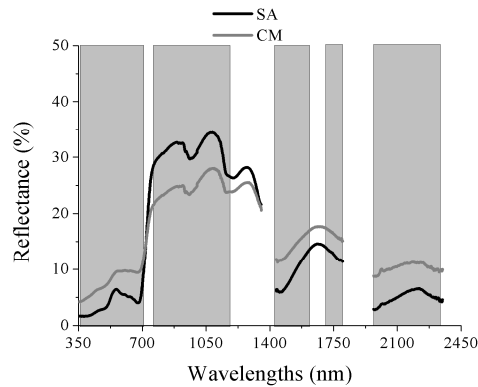


Fig.1. ANOVA Results for each class pair SA vs. PA, SA vs. KC, and SA vs. CM. The grey areas show the wavebands where there are significant differences between each class pairs within the electromagnetic spectrum.

3.2 CART results

Table 2 showed the optimal minimum number of wavelengths (primary splitters) that discriminate *S. alterniflora* from the other three native species, using CART analysis. The selection of the optimal wavelengths was done for each individual class pair (SA vs. PA, SA vs. CM, SA vs. KC), in each spectral region. The most number of primary splitters was obtained for SA vs. KC, while the smallest number was obtained for SA vs. KC. These bands were spread in the whole spectral range, but were frequently located in the near infrared.

Table 2

Wavelengths (nm) selected by CART (primary splitters) for each individual class pair, in each spectral region.

	SA vs. PA	SA vs. KC	SA vs. CM
Blue (350-449)	406	-	396
Green (450-549)	-	-	450, 526
Red (550-649)	-	-	550, 649
Red edge (650-749)	-	714, 718, 731	669, 740
Near infrared (750-1299)	959, 1001, 1015, 1127, 1128, 1139, 1156, 1225	1001, 1119, 1196, 1250	750, 836, 877, 952, 1001
Mid infrared (1300-2349)	1448, 1454, 1467	1300, 1341	1444, 1970, 1971

Table 3 showed the complexity of CART models and the misclassification result. The CART classification accuracy for SA vs. PA was the lowest, while the highest classification accuracy was obtained for SA vs. KC. The classification trees in the near infrared region were relatively more complex and higher classification accuracy compared with the other spectral regions. Likewise, the number of wavelengths identified by the CART models in the near infrared was much more.

Table 3

Tree sequences for CART model and the misclassification rate in each spectral region

	Spectral region	Number nodes	Number terminal nodes	Tree depth	Misclassification
SA vs. PA	Blue	3	2	1	0.168
	Near infrared	17	9	5	0.021
	Mid infrared	7	4	3	0.031
	Average	9	5	3.67	0.073
SA vs. KC	Red edge	7	4	3	0.029
	Near infrared	7	4	3	0.014
	Mid infrared	11	6	3	0.014
	Average	8.3	4.6	3	0.019
SA vs. CM	Blue	3	2	1	0.039
	Green	7	4	3	0.026
	Red	7	4	3	0.039
	Red edge	5	3	2	0.066
	Near infrared	13	7	4	0.026
	Mid infrared	7	4	3	0
	Average	7	4	2.6	0.032

3.3 Spectral separability results

The separability index (JM distance) was calculated between each individual class pair using the optimal minimum subset of bands selected by CART (Table 4). The results showed that separability between all of the individual pairs was greater than the acceptable JM distance threshold of 1.372, and reached the maximum value 1.414. In other words, the spectral separability is very good between *S. alterniflora* and the three native species, which selected by CART.

Table 4

The values of JM distance for each individual class pair within the selected optimal band

	SA vs. PA	SA vs. KC	SA vs. CM
JM	1.414	1.414	1.414

4. Discussion

Field spectrometry has been widely used for species discrimination^[21], but litter attention has received in discriminating coastal wetland plants. The visual analysis of the mean reflectance of different wetland vegetation species showed a lot of overlap, but there still had some wavelengths showed crossing-over (Fig.1) due to the variation of spectral. This study selected those bands that were statistically significant, allowing *S. alterniflora* to be discriminated from the other three native species through ANOVA with the post-hoc Scheffe test. CART analysis was further applied

to reduce the number of significant bands selected by ANOVA analysis to fewer bands that could optimally discriminate the *S. alterniflora* from the other three native species. Finally, we used JM distance to assess the separability of the selected optimal wavelengths. Results of this study show that the discrimination of *S. alterniflora* from native species is possible at the field level using field spectrometry, which would be scaling up to airborne remote sensing.

4.1 Spectral discrimination of *S. alterniflora* and spectral variation

In this study, we found that *S. alterniflora* was spectrally separable from the other three native species. Wavelength located in the near infrared and red edge regions were recurrently selected in the discrimination of *S. alterniflora* from the native species. This confirms the results of previous studies that state wetland vegetation have greatest variation in the near infrared regions and near infrared regions [22, 23]. In the red edge region, canopy reflectance is strongly correlated with plant biochemical and biophysical parameters, especially pigment concentration, nitrogen concentration, and water content [24, 25]. Although these parameters were not directly measured in this study, it is likely that the variation in the red edge of wetland plants due to the variation on these parameters. In the near infrared, vegetation types generally have a high reflectance and transmittance, very low absorption [4]. The physical control is internal leaf structures [26]. Thus, variation in the near infrared between *S. alterniflora* and the native species may be due to different canopy structure.

However, our study also proved the optimal bands for discrimination of the *S. alterniflora* in the mid infrared region, which revealed a high variability reflectance in this region. Previous studies showed that, the spectral variability in the mid infrared region is mainly due to strong water absorption and minor absorption of biochemical content [27]. Unlike the land-plant, *S. alterniflora* and the other wetland plants are basically restricted to the area that is permanently either wet or flooded throughout the year. This result in a higher water content in canopies compared to other species. Thus, variation in the mid infrared between *S. alterniflora* and native species may be due to leaf water content or unique hydrologic environment. Similar results were also found by Everitt et al. [28] and Fernandes et al [14].

4.2 Band selection using CART

We used CART to select the optimal band to discriminate *S. alterniflora* from native species, and JM distance values reached the maximum of 1.414. Table 2 and Table 3 showed the bands selected and the misclassification error rate. Relative to other studies, the classification result was very good [29]. Table 3 also revealed that classification accuracy used the bands selected by CART in the near infrared and in the mid infrared relatively higher than other band, especially in the near infrared, which can guide us to choose the appropriate sensor to identify *S. alterniflora*. That's to say, band selection in this study has the possibility exists to extend to airborne and space borne level.

5. Conclusion

The following conclusions can be drawn based on the results of the present study:

1. Field spectrometer measurements at canopy level can be used to discriminate *S. alterniflora* from *P. australis*, *C. malaccensis* and *K. candel*. This implies that the mean spectral reflectance of *S. alterniflora* is different from the other three native species in Min river wetland.

2. Classification and regression trees (CART) can be used to reduce considerably the dimensionality and to select the most important bands for discriminating *S. alterniflora* from the other three species with a high classification accuracy.

3. The use of CART has revealed that the greatest discrimination power for *S. alterniflora* is located in the near infrared, in the mid infrared, and red edge regions, especially in the near infrared regions.

Overall, the outcome of this study is that the possibility exists to extend this work to airborne and satellite hyperspectral sensor data for discriminating *S. alterniflora* in Min river wetland.

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